24AD503 COMPUTER NETWORKS

UNIT I – INTRODUCTION AND PHYSICAL LAYER

Data communication fundamentals, Network criteria such as performance, reliability, and security. Types of networks - OSI reference model and TCP/IP model with comparison. Transmission media, Switching techniques, Performance metrics including bandwidth, throughput, delay, and latency.

UNIT II – DATA LINK LAYER

Framing techniques - Error detection techniques - Error correction - Flow control mechanisms including Stop-and Wait and Sliding Window protocols. Automatic Repeat Request (ARQ) protocols including Go- Back-N and Selective Repeat. Multiple access protocols such as ALOHA and CSMA/CD. Ethernet frame structure and addressing.

UNIT III – NETWORK LAYER

IPv4 addressing and IPv6 addressing concepts. Subnetting and supernetting using CIDR. Routing algorithms including Distance Vector and Link State - Overview of routing protocols such as RIP and OSPF – Congestion control techniques - ICMP protocol and its message types.

UNIT IV – TRANSPORT LAYER

Transport layer services - Multiplexing and demultiplexing - UDP protocol features and header format – TCP protocol - segment structure, connection establishment, flow control, and congestion control algorithms and congestion avoidance. Port addressing and socket programming basics - Quality of Service (QoS) parameters.

UNIT V – APPLICATION LAYER AND WIRELESS NETWORK

Application layer protocols - HTTP, FTP, SMTP, and DNS- Wireless networks: WLAN, Bluetooth, Wi-Fi architecture.

TEXT BOOKS

1. Behrouz A. Forouzan, Data Communications and Networking, 6 th Edition, McGraw Hill Education, 2022

2. James F. Kurose and Keith W. Ross, Computer Networking: A Top-Down Approach, 8 th Edition, Pearson Education, 2022

REFERENCES:

1. Andrew S. Tanenbaum and David J. Wetherall, Computer Networks, 6th Edition, Pearson Education, 2021.

2. William Stallings, Data and Computer Communications, 10th Edition, Pearson Education, 2013.